Laboratory Manual

For

Electrical Drives and Instrumentation Sessional
EEE-2422

Department of Computer Science and Engineering

IIUC

Experiment No.	Name of the experiment
01	Familiarization with instruments used in Electrical Drives Lab.
02	Introduction to Arduino Uno R3.
03	Setting up the Arduino IDE.
04	Familiarization with different types of sensor.
05	Familiarization with different display elements.
05	Familiarization with output devices for Arduino Uno R3
06	Ultrasonic Sensor.
07	Arduino project-I (Blinking LED light)
08	Arduino with Simulation Software. (Blinking LED Traffic lights)
09	Circuit Simulation Using Proteus
	Lab Projects

Course Code: EEE-2422	
Course Title: Electrical Drives and Instrumentation Sessional.	
Credit Hours: 1	Contact Hours: 2 per week

Reference Books:

1. **A textbook of Electrical Technology by B.L Thereja & A.K. Thereja**
2. **Electric Machinery Fundamentals by Stephen J. Chapman**

Introduction

The purpose of an Electrical Instrumentation with Arduino Lab is to provide students with hands-on experience in designing and building electronic circuits and systems using the Arduino platform. Arduino is an open-source platform that allows users to easily create interactive electronic projects.

In this lab, students learn the basic principles of electronics, including circuit design, component selection, and measurement techniques. They also learn how to use Arduino boards and programming languages like C to control and monitor various sensors and actuators. By the end of the lab, students should be able to design and implement basic electronic systems using Arduino boards, write and debug programs in C, and effectively use various sensors and measurement tools.

The use of Arduino in electrical drives is intended to offer a versatile and affordable platform for controlling and monitoring various electrical drive parameters. From small motors to huge industrial machines, electrical drives are utilized in a variety of industrial applications.

Designing and implementing control systems for electrical drives is made simple and economical by the open-source platform Arduino. It permits the integration of several sensors and actuators, including power switches, current sensors, and encoders, which are used to monitor and manage the operation of the drive system.

Experiment No. 1

Name of the Experiment: Familiarization with the instruments used in Electrical Drives Lab.

Objective: Familiarization with the instruments used in Electrical Drives lab.

Equipments:

1.	Transformer
2.	DC motor
3.	Induction Motor
4.	Inductive Load
5.	Resistive Load
6.	DC generator
7.	DC power supply
8.	Capacitive Load
9.	Synchronous generator
10.	AC power Supply
11.	Synchronous Motor
12.	RPM meter.

Report: Write a report including

a. Application of the instruments.
b. Circuit Diagram of the instruments.
c. Ratings of the Instruments.

Report: The descriptions of the instruments used in the Electrical Drives lab are as follows:
1. Transformer:

[write a short description of the equipment including
a. Definition
b. Ratings
c. Application
d. Add a figure/circuit diagram]

2. ....
....

Discussion:

Experiment No. 2

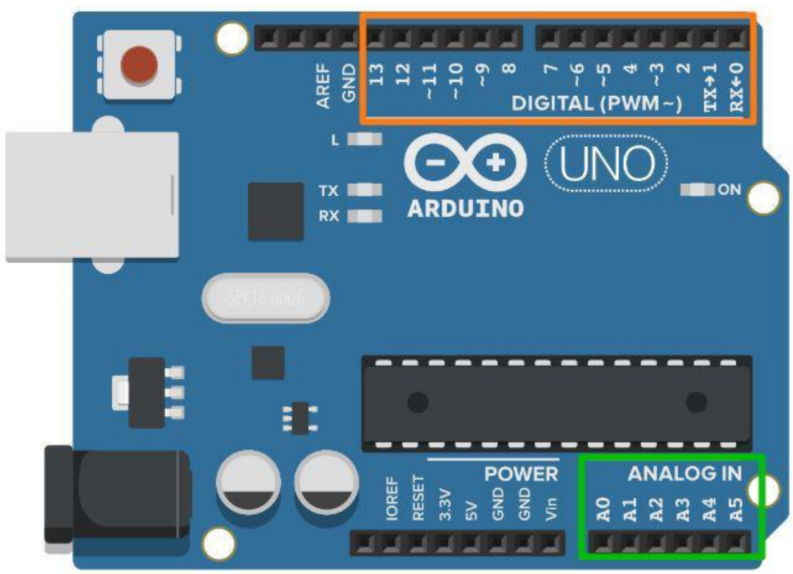
Name of the Experiment: Introduction to Arduino Uno R3

Objective: To know about Arduino Uno R3.

Equipment:

- Arduino board (better to choose Arduino Uno, but Nano and Mega will also do).
- Arduino Software (IDE)
- USB loader.

Arduino Pins:



On Arduino Uno, you get:

14 digital pins (orange) ,6 analog pins (green)

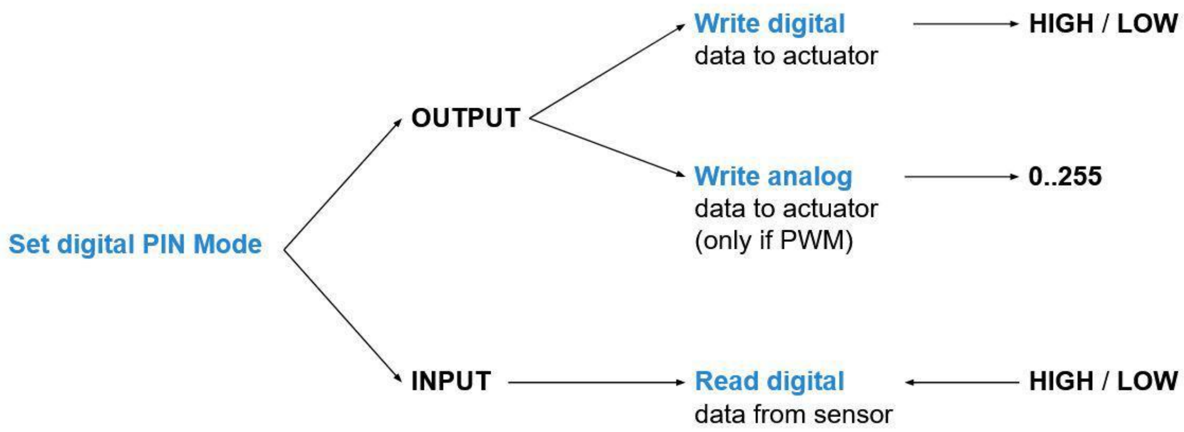
Digital Pins

Digital pins are used to read/write digital values (which are binary values: HIGH/LOW or 1/0).

For any given digital pin on the Arduino, you can decide to set it as INPUT or OUTPUT, with the pinMode(pin_number, mode)function.

Then:

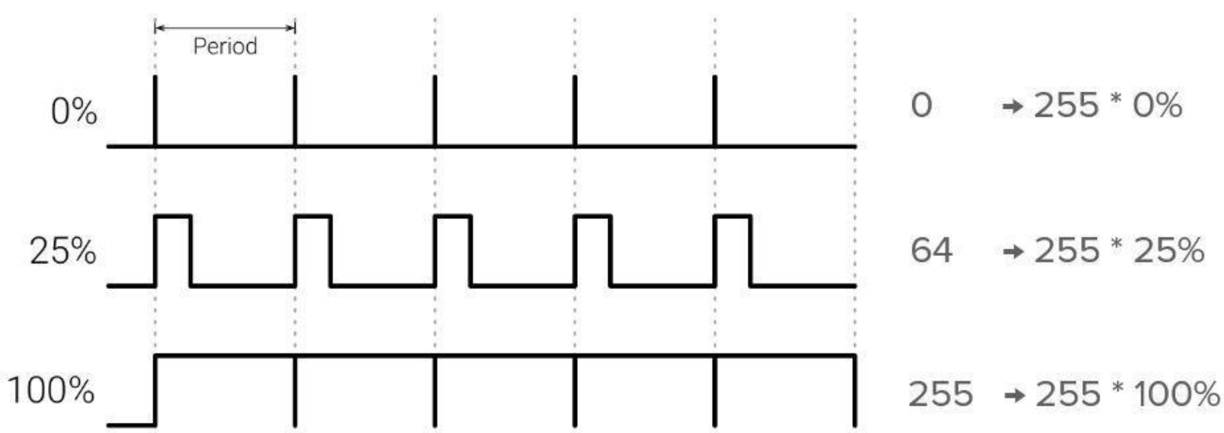
- For INPUT pins, you can read the state of the pin with *digitalRead(pin _number)*
- For OUTPUT pins, you can write the state of the pin with *digitalWrite(pin _number)*



Note:

- For some of the digital pins (with a “~” symbol next to the number on the board), you can

  use a PWM with the function *analogWrite(pin_number, value between 0..255).*



Analog Pins

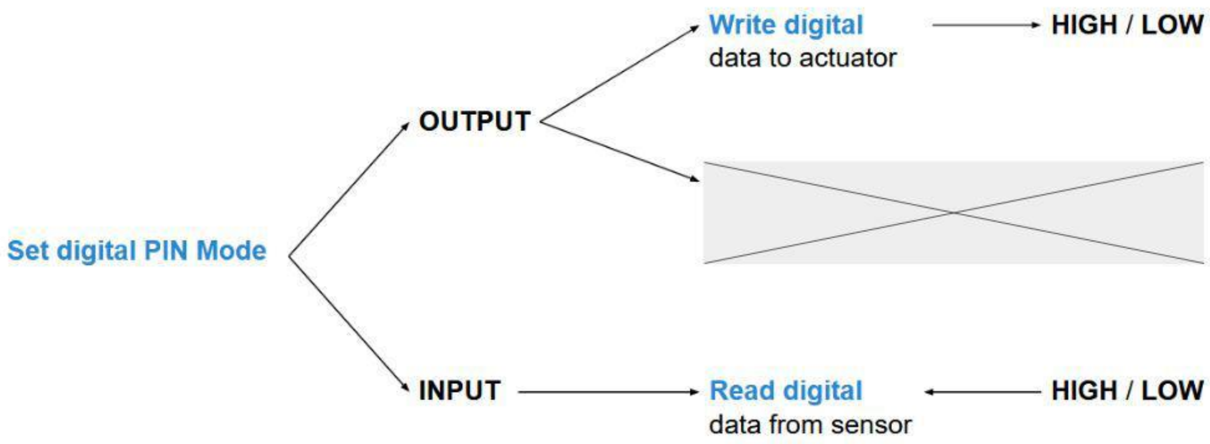
Analog pins are used to read analog values (variation of the voltage between 0 and 5V).

You don’t need to set up the mode for an analog input pin, it’s already set as INPUT.

Then: To read an analog value, you can use analogRead(pin_number), which will convert the analog signal into a digital value between 0 and 1023, that you can use in your code.

Or:

- You can use any analog pin as a standard digital pin. Very handy when you've already used all digital pins. To do that, don't forget to set the mode for the pin with *pinMode(pin_number, mode)*. Note that you can't use the PWM functionality on analog pins.



Experiment-3:

Name of the Experiment: Setting up the Arduino IDE.

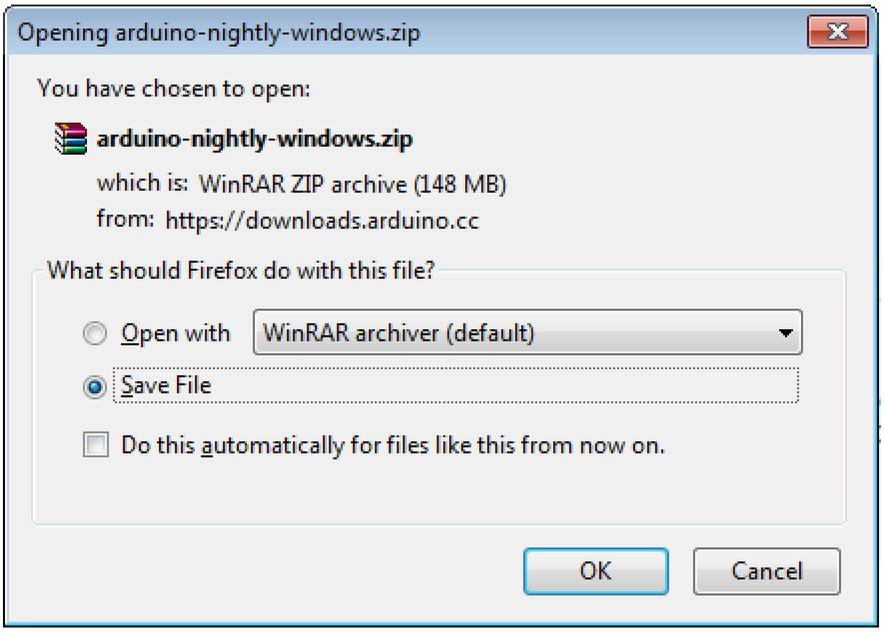
Objectives: will learn in easy steps, how to set up the Arduino IDE on our computer and prepare the board to receive the program via USB cable.

Step 1 − First you must have your Arduino board (you can choose your favorite board) and a USB cable. In case you use Arduino UNO, Arduino Duemilanove, Nano, Arduino Mega 2560, or Diecimila, you will need a standard USB cable (A plug to B plug), the kind you would connect to a USB printer as shown in the following image.

In case you use Arduino Nano, you will need an A to Mini-B cable instead as shown in the following image.

Step 2 − Download Arduino IDE Software.

You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.



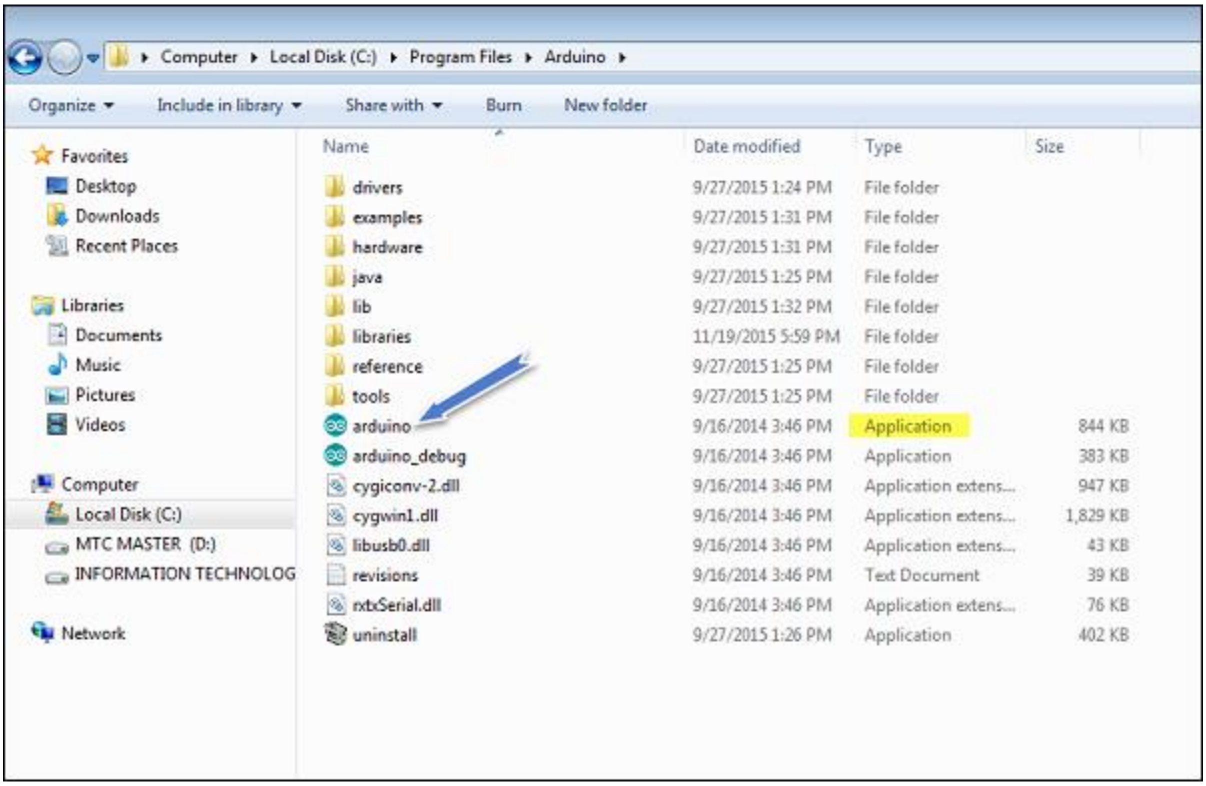
Step 3 − Power up your board.

The Arduino Uno, Mega, Duemilanove and Arduino Nano automatically draw power from either, the USB connection to the computer or an external power supply. If you are using an Arduino Diecimila, you have to make sure that the board is configured to draw power from the USB connection. The power source is selected with a jumper, a small piece of plastic that fits onto two of the three pins between the USB and power jacks. Check that it is on the two pins closest to the USB port.

Connect the Arduino board to your computer using the USB cable. The green power LED (labeled PWR) should glow.

Step 4 − Launch Arduino IDE.

After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Double-click the icon to start the IDE.

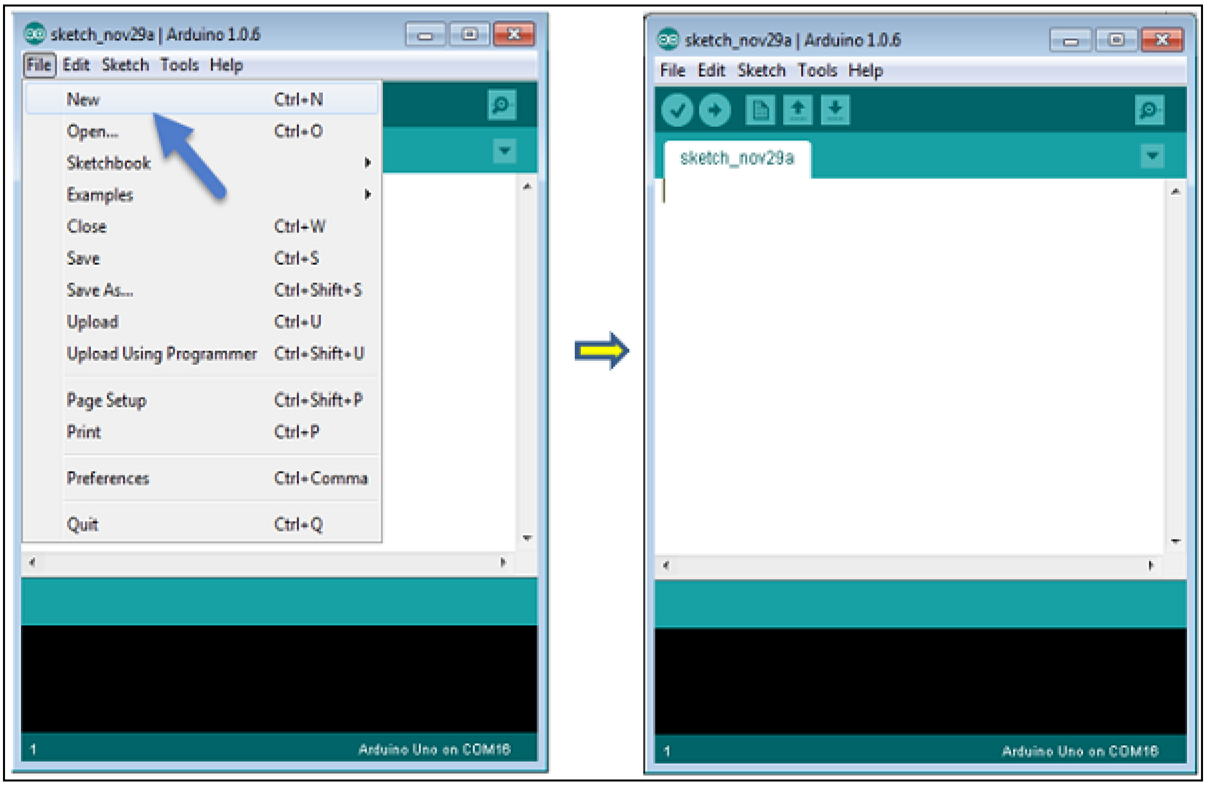


Step 5 − Open your first project.

Once the software starts, you have two options −

- Create a new project.
- Open an existing project example.

To create a new project, select File → **New**.



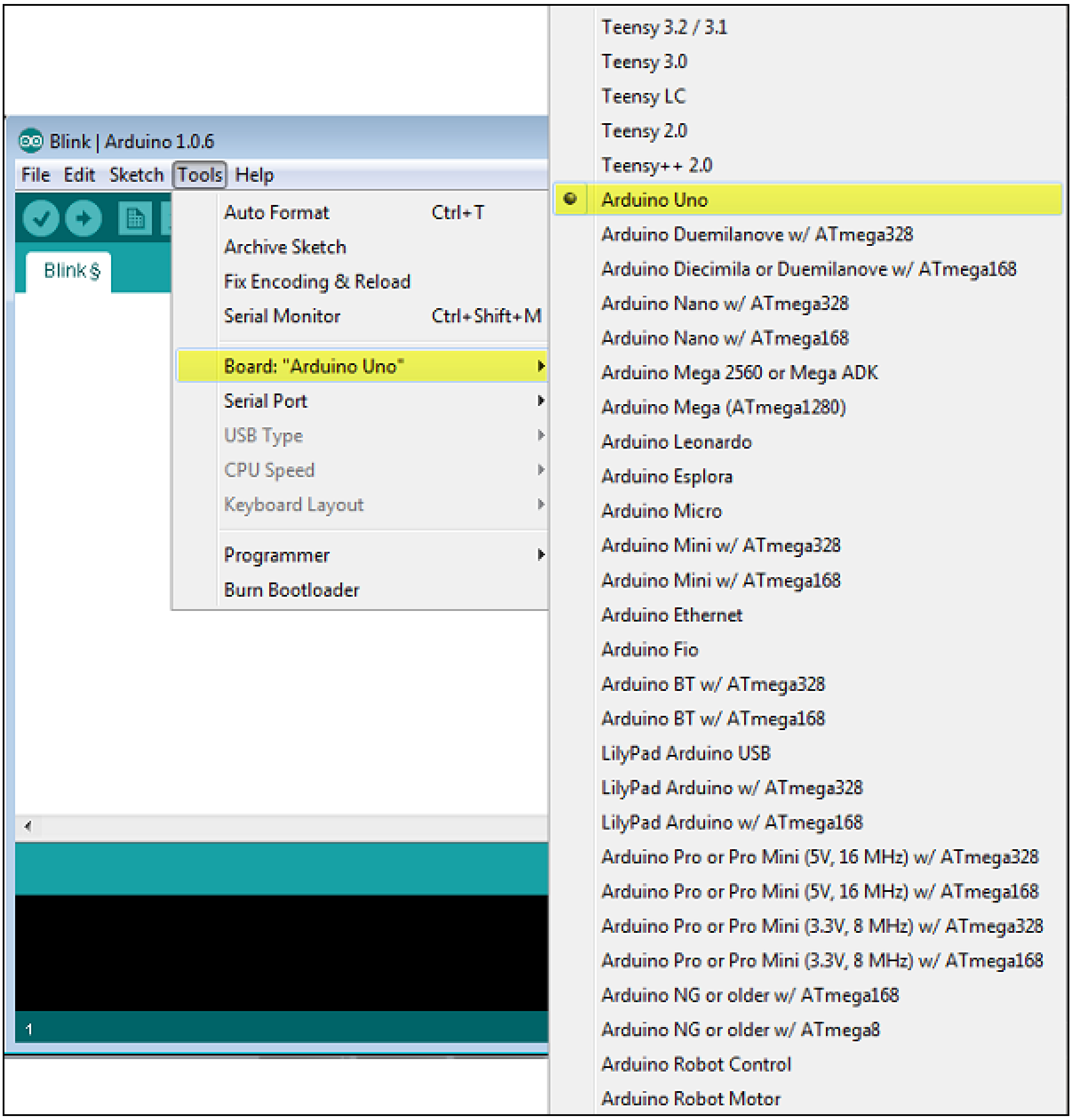
To open an existing project example, select File → Example → Basics → Blink.

Here, we are selecting just one of the examples with the name **Blink**. It turns the LED on and off with some time delay. You can select any other example from the list.

Step 6 − Select your Arduino board.

To avoid any error while uploading your program to the board, you must select the correct Arduino board name, which matches with the board connected to your computer.

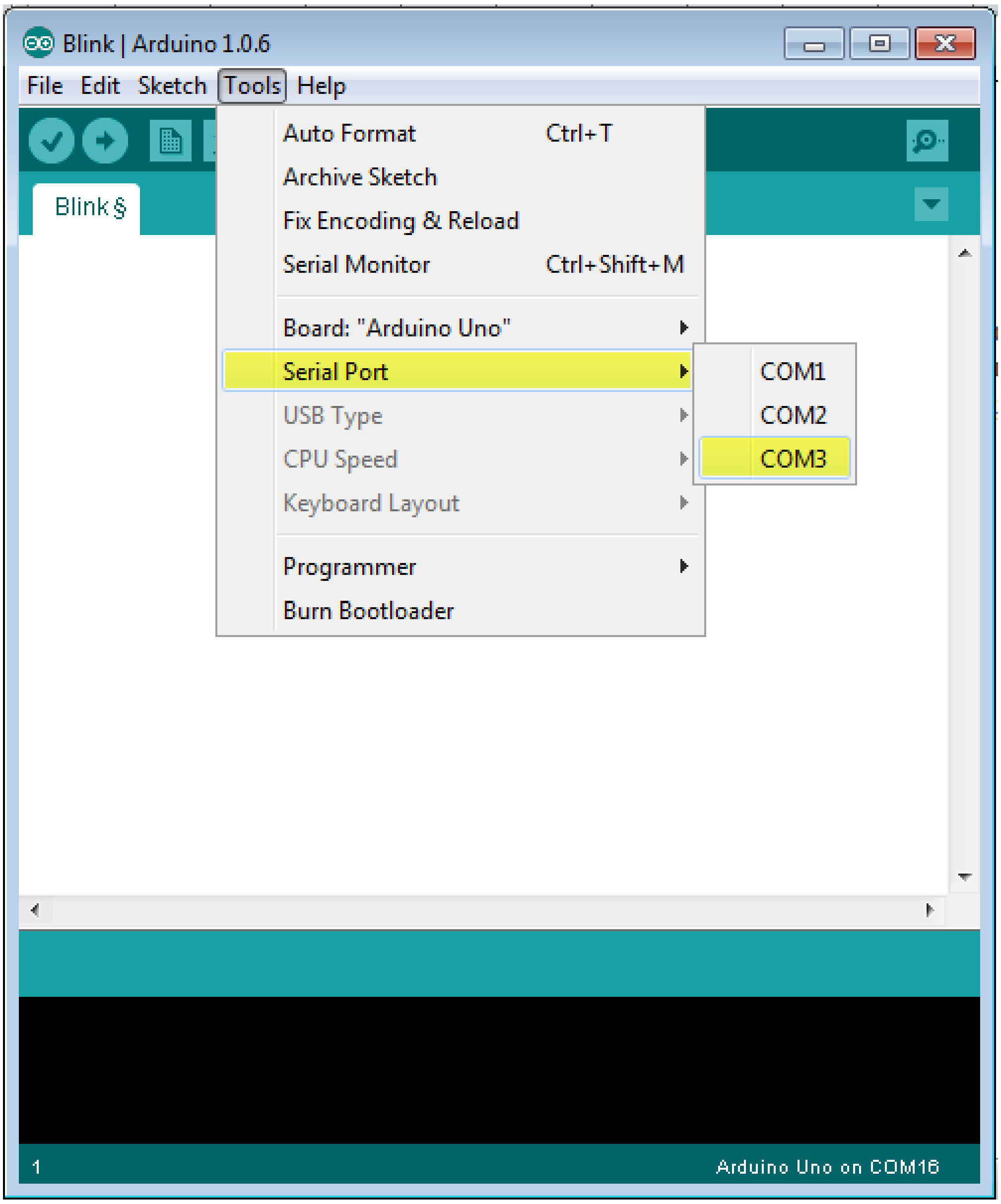
Go to Tools → Board and select your board.



Here, we have selected Arduino Uno board according to our tutorial, but you must select the name matching the board that you are using.

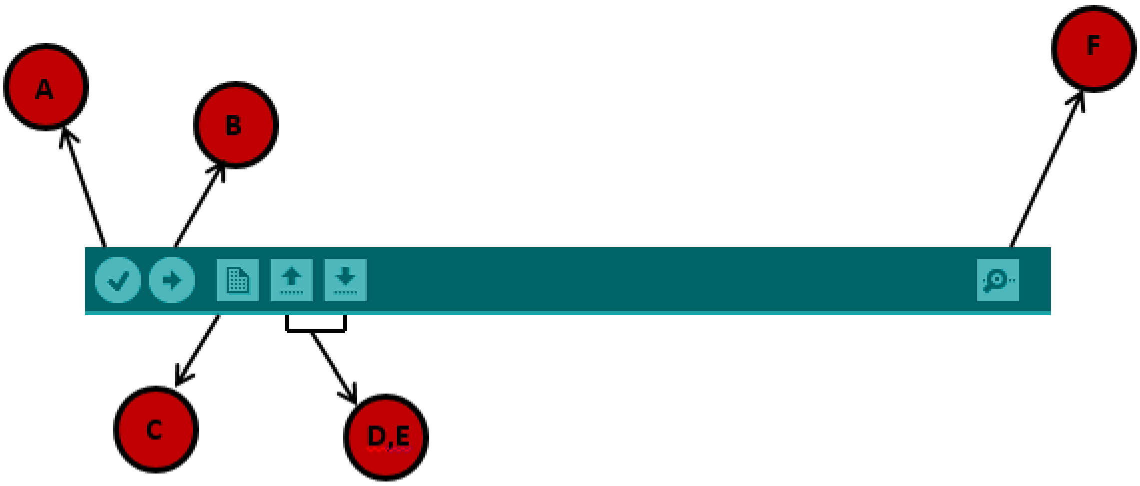
Step 7 − Select your serial port.

Select the serial device of the Arduino board. Go to **Tools → Serial Port** menu. This is likely to be COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports). To find out, you can disconnect your Arduino board and re-open the menu, the entry that disappears should be of the Arduino board. Reconnect the board and select that serial port.



Step 8 − Upload the program to your board.

Before explaining how we can upload our program to the board, we must demonstrate the function of each symbol appearing in the Arduino IDE toolbar.



A − Used to check if there is any compilation error.

B − Used to upload a program to the Arduino board.

C − Shortcut used to create a new sketch.

D − Used to directly open one of the example sketch.

E − Used to save your sketch.

F − Serial monitor used to receive serial data from the board and send the serial data to the board.

Now, simply click the "Upload" button in the environment. Wait a few seconds; you will see the RX and TX LEDs on the board, flashing. If the upload is successful, the message "Done uploading" will appear in the status bar.

Note − If you have an Arduino Mini, NG, or other board, you need to press the reset button physically on the board, immediately before clicking the upload button on the Arduino Software.

Experiment-4: ***Exp-2***

Name of the Experiment: Familiarization with different types of sensor.

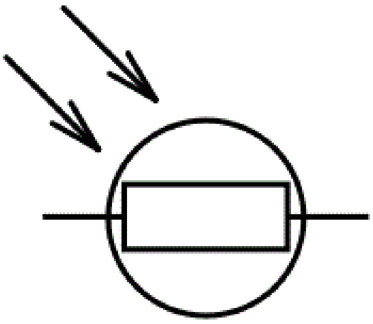
Objectives:

1. Familiarization with LDR as a sensor device for measurement.
2. Familiarization with thermistor as a sensing device for measurement.
3. Familiarization with Temperature sensor LM35 as a sensing device for measurement.
4. Familiarization with Gas sensor MQ4 as a sensing device for measurement.

LDR (Light Dependent Resistor)

A photo resistor (or light-dependent resistor, LDR, or photocell) is a light-controlled variable resistor. The resistance of a photo resistor decreases with increasing incident light intensity; in other words, it exhibits photoconductivity.

They are also called as photo conductors, photo conductive cells or simply photocells. They are made up of semiconductor materials having high resistance. There are many different symbols used to indicate a LDR, one of the most commonly used symbol is shown in the figure below. The arrow indicates light falling on it.



Apparatus:
Report:
1. LDR
[Definition, Working Principle, Symbol, Characteristic Curve, Application]
Discussion:

Figure 1.1: Symbol of LDR

Types of Light Dependent Resistors:

Based on the materials used they are classified as: i) Intrinsic photo resistors (Un doped semiconductor): These are made of pure semiconductor materials such as silicon or germanium. Electrons get excited from valance band to conduction band when photons of enough energy fall on it and number charge carriers is increased.

Extrinsic photo resistors: These are semiconductor materials doped with impurities which are called as dopants. Theses dopants create new energy bands above the valence band which are filled with electrons. Hence this reduces the band gap and less energy is required in exciting them. Extrinsic photo resistors is generally used for long wavelengths.

Construction of a Photocell:

The structure of a light dependent resistor consists of a light sensitive material which is deposited on an insulating substrate such as ceramic. The material is deposited in zigzag pattern in order to obtain the desired resistance & power rating. This zigzag area separates the metal deposited areas into two regions. Then the ohmic contacts are made on the either sides of the area. The resistances of these contacts should be as less as possible to make sure that the resistance mainly changes due to the effect of light only. Materials normally used are cadmium sulphide, cadmium selenide, indium antimonide and cadmium sulphonide. The use of lead and cadmium is avoided as they are harmful to the environment.

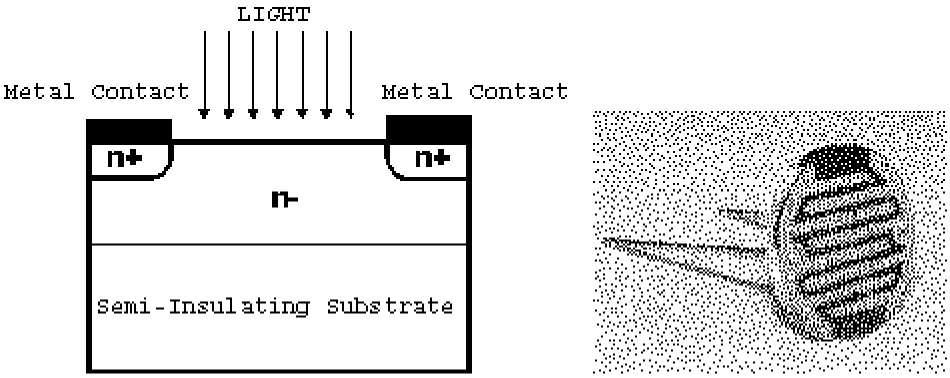


Figure 1.2: Construction of LDR

Working Principle of LDR:

A light dependent resistor works on the principle of photo conductivity. Photo conductivity is an optical phenomenon in which the materials conductivity is increased when light is absorbed by the material. When light falls i.e. when the photons fall on the device, the electrons in the valence band of the semiconductor material are excited to the conduction band. These photons in the incident light should have energy greater than the band gap of the semiconductor material to make the electrons jump from the valence band to the conduction band. Hence when light having enough energy strikes on the device, more and more electrons are excited to the conduction band which results in large number of charge carriers. The result of this process is more and more current starts flowing through the device when the circuit is closed and hence it is said that the resistance of the device has been decreased. This is the most common working principle of LDR.

Characteristics of LDR:

LDR's are light dependent devices whose resistance is decreased when light falls on them and that is increased in the dark. When a light dependent resistor is kept in dark, its resistance is very high. This resistance is called as dark resistance. It can be as high as 1012 Ω and if the device is allowed to absorb light its resistance will be decreased drastically. If a constant voltage is applied to it and intensity of light is increased the current starts increasing. Figure below shows resistance vs. illumination curve for a particular LDR.

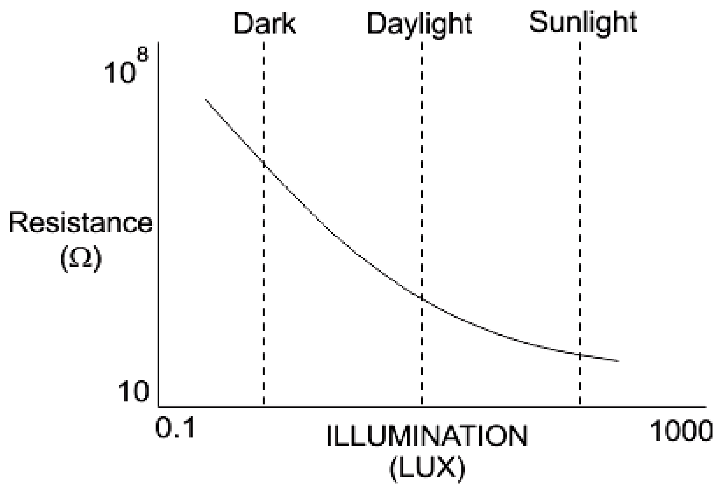


Figure 1.3: Characteristics curve of LDR

Photocells or LDR’s are nonlinear devices. There sensitivity varies with the wavelength of light incident on them. Some photocells might not at all response to a certain range of wavelengths. Based on the material used different cells have different spectral response curves.

When light is incident on a photocell it usually takes about 8 to 12ms for the change in resistance to take place, while it takes one or more seconds for the resistance to rise back again to its initial value after removal of light. This phenomenon is called as resistance recovery rate. This property is used in audio compressors. Also, LDR's are less sensitive than photo diodes and photo transistor. (A photo diode and a photocell (LDR) are not the same, a photo-diode is a p-n junction semiconductor device that converts light to electricity, whereas a photocell is a passive device, there is no p-n junction in this nor it "converts" light to electricity).

Applications of LDR:

LDR's have low cost and simple structure.

They are often used as light sensors. They are used when there is a need to detect absences or presences of light like in a camera light meter.

Used in street lamps, alarm clock, burglar alarm circuits, light intensity meters, for counting the packages moving on a conveyor belt, etc.

2) Thermistors

A thermistor is a temperature sensor constructed of semiconductor material that exhibits a large modification in resistance in proportion to a tiny low modification in temperature. Thermostats are inexpensive, rugged, reliable and responds quickly. Because of these qualities thermistors are used to measure simple temperature measurements, but not for high temperatures. Thermistor is easy to use, cheap, durable and respond predictably to a change in temperature. Thermistors are mostly used in digital thermometers and home appliances such as refrigerator, ovens, and so on. Stability, sensitivity and time constant are the final properties of thermistor that create these thermistors sturdy, portable, cost-efficient, sensitive and best to measure single-point temperature.

Materials used for Thermistors and their Forms:

The thermistors are made up of ceramic like semiconducting materials. They are mostly composed of oxides of manganese, nickel and cobalt having the resistivity's if about 100 to 450,000 ohm-cm. Since the resistivity of the thermistors is very high the resistance of the circuit in which they are connected for measurement of temperature can be measured easily. This resistance is calibrated against, the input quantity, which is the temperature, and its value can be obtained easily.

Thermistors are available in various shapes like disc, rod, washer, bead etc. They are of small size and they all can be fitted easily to the body whose temperature has to be measured and also can be connected to the circuit easily. Most of the thermistors are quite cheap.

Figure 1.4: Constructional shape of Thermistor.

Working Principle of Thermistors:

As mentioned earlier the resistance of the thermistors decreases with the increase its temperature. The resistance of thermistor is given by:

R = Ro ek

K = β (1/T – 1/To)

Where R is the resistance of the thermistor at any temperature T in oK (degree Kelvin)

Ro is the resistance of the thermistors at particular reference temperature Toin oK e is the base of the Naperian logarithms

β is a constant whose value ranges from 3400 to 3900 depending on the material used for the thermistors and its composition.

The thermistor acts as the temperature sensor and it is placed on the body whose temperature is to be measured. It is also connected in the electric circuit. When the temperature of the body changes, the resistance of the thermistor also changes, which is indicated by the circuit directly as the temperature since resistance is calibrated against the temperature. The thermistor can also be used for some control which is dependent on the temperature.

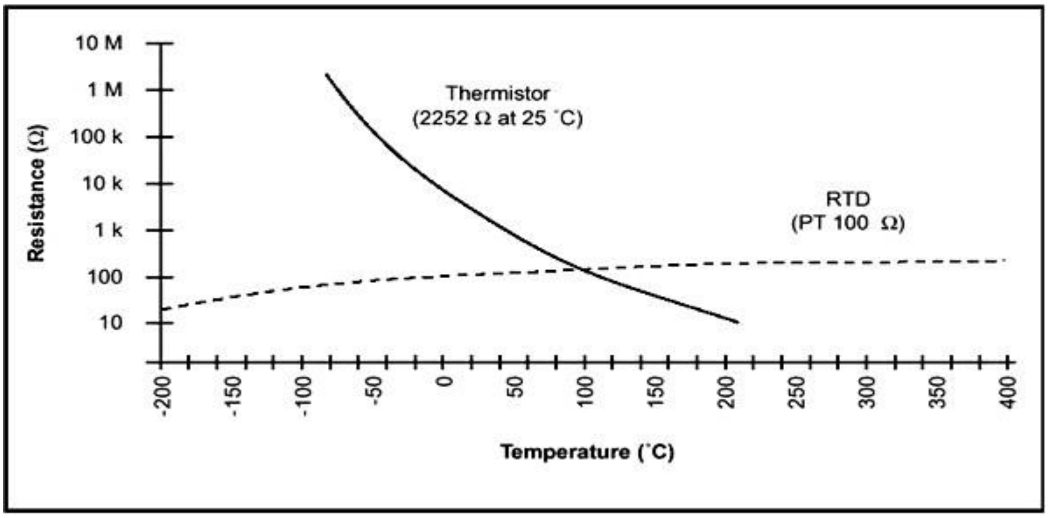


Figure 1.5: Characteristics Curve of Thermistor.

RTD: A Resistance Thermometer or Resistance Temperature Detector is a device which used to determine the temperature by measuring the resistance of pure electrical wire. This wire is referred to as a temperature sensor. If we want to measure temperature with high accuracy, RTD is the only one solution in industries.

Application and Advantages of Thermistor:

When the resistors are connected in the electrical circuit, heat is dissipated in the circuit due to flow of current. This heat tends to increase the temperature of the resistor due to which their resistance changes. For the thermistor the definite value of the resistance is reached at the given ambient conditions due to which the effect of this heat is reduced.

In certain cases even the ambient conditions keep on changing, this is compensated by the negative temperature characteristics of the thermistor. This is quite convenient against the materials that have positive resistance characteristics for the temperature.

The thermistors are used not only for the measurement of temperature, but also for the measurement of pressure, liquid level, power etc.

They are also used as the controls, overload protectors, giving warnings etc.

The size of the thermistors is very small and they are very low in cost. However, since their size is small they have to be operated at lower current levels.

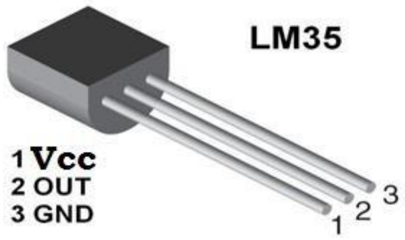
3) Temperature Sensor (LM35)

The LM35-series devices are precision integrated-circuit temperature sensors, with an output voltage linearly proportional to the Centigrade temperature. The LM35 device has an advantage over linear temperature sensors calibrated in Kelvin, as the user is not required to subtract a large constant voltage from the output to obtain convenient Centigrade scaling. The LM35 device does not require any external calibration or trimming to provide typical accuracies of ± ¼ °C at room temperature and ± ¾

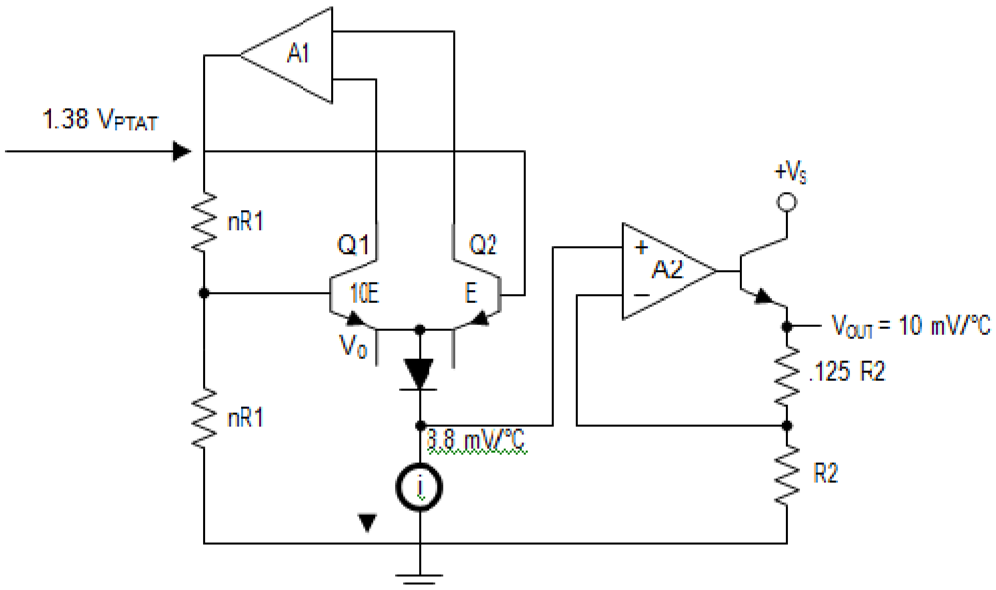
°C over a full −55°C to 150°C temperature range. Lower cost is assured by trimming and calibration at the wafer level. The low output impedance, linear output, and precise inherent calibration of the LM35 device makes interfacing to readout or control circuitry especially easy. The device is used with single power supplies, or with plus and minus supplies. As the LM35 device draws only 60 μA from the supply, it has very low self-heating of less than 0.1°C in still air. The LM35 device is rated to operate over a −55°C to 150°C temperature range, while the LM35C device is rated for a −40°C to 110°C range (−10° with improved accuracy). The temperature-sensing element is comprised of a delta- V BE architecture.

The temperature-sensing element is then buffered by an amplifier and provided to the VOUT pin. The amplifier has a simple class A output stage with typical 0.5-Ω output impedance as shown in the *Functional Block Diagram*. Therefore, the LM35 can only source current and its sinking capability is limited to 1 μA.

Pin Description:



Pin No	Function	Name
1	Supply voltage; 5V (+35V to -2V)	Vcc
2	Output voltage (+6V to -1V)	Output
3	Ground (0V)	Ground



Functional Block Diagram:

Applications:

- Power Supplies
- Appliances
- HVAC
- Battery Management

4) Gas Sensor

In current technology scenario, monitoring of gases produced is very important. From home appliances, such as air conditioners to electric chimneys and safety systems at industries monitoring of gases is very crucial. Gas sensors are very important part of such systems. Small like a nose, gas sensors spontaneously react to the gas present, thus keeping the system updated about any alterations that occur in the concentration of molecules at gaseous state.

Construction & Working Principle:

Gas sensors are available in wide specifications depending on the sensitivity levels, type of gas to be sensed, physical dimensions and numerous other factors. This Insight covers a methane gas sensor that can sense gases such as ammonia which might get produced from methane. When a gas interacts with this sensor, it is first ionized into its constituents and is then adsorbed by the sensing element. This adsorption creates a potential difference on the element which is conveyed to the processor unit through output pins in form of current.

The gas sensor module consists of a steel exoskeleton under which a sensing element is housed. This sensing element is subjected to current through connecting leads. This current is known as heating current through it, the gases coming close to the sensing element get ionized and are absorbed by the sensing element. This changes the resistance of the sensing element which alters the value of the current going out of it.

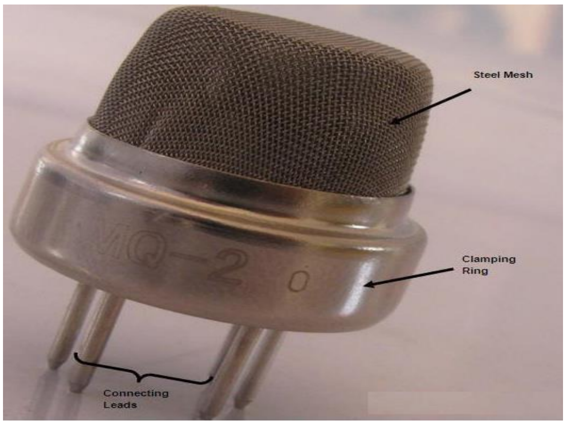


Figure 1.6: Gas Sensor

Figure 1.7: Gas Sensor (Internal)

The top of the gas sensor is removed off to see the internals parts of the sensor: sensing element and connection wiring. The hexapod structure is constituted by the sensing element and six connecting legs that extend beyond the Bakelite base. Fig. shows the hollow sensing element which is made up from Aluminum Oxide based ceramic and has a coating of tin oxide. Using a ceramic substrate increases the heating efficiency and tin oxide, being sensitive towards adsorbing desired gas’ components (in this case methane and its products) suffices as sensing coating.

The leads responsible for heating the sensing element are connected through Nickel-Chromium, well known conductive alloy. Leads responsible for output signals are connected using platinum wires which convey small changes in the current that passes through the sensing element. The platinum wires are connected to the body of the sensing element while Nickel-Chromium wires pass through its hollow structure.

Gas Sensor - MQ-4:

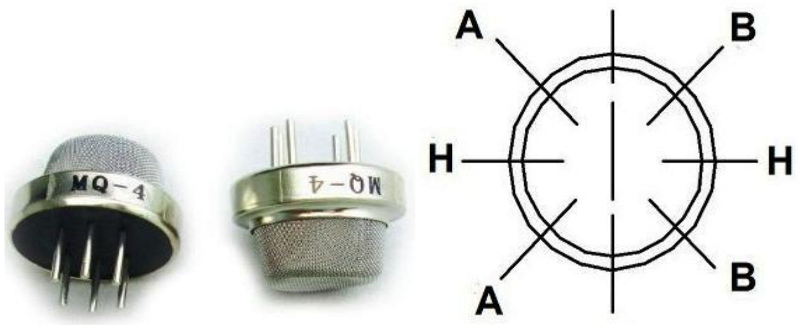
Description:

This is a simple-to-use compressed natural gas (CNG)sensor, suitable for sensing natural gas (composed of mostly Methane [CH4]) concentrations in the air. The MQ-4 can detect natural gas concentrations anywhere from 200 to 10000ppm.

Sensitive material of MQ-4 gas sensor is SnO2, which with lower conductivity in clean air. When the target combustible gas exists, the sensor's conductivity is higher along with the gas concentration rising. MQ-4 gas sensor has high sensitivity to Methane, also to Propane and Butane. The sensor could be used to detect different combustible gas, especially Methane, it is with low cost and suitable for different application.

This sensor has a high sensitivity and fast response time. The sensor's output is an analog resistance. The drive circuit is very simple; all you need to do is power the heater coil with 5V, add a load resistance, and connect the output to an ADC.

Configuration:



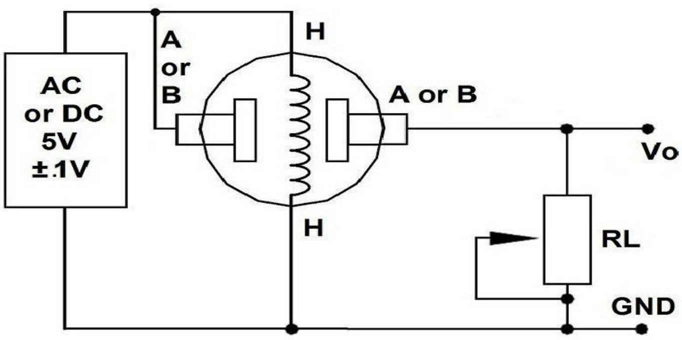


Fig-1.8: Internal Diagram

Characteristics:

- Good sensitivity to Combustible gas in wide range
- High sensitivity to Natural gas
- Long life and low cost
- Simple drive circuit
- Applications:
- Domestic gas leakage detector
- Industrial Combustible gas detector
- Portable gas detector.

Experiment-5: Familiarization with different display elements.

Objectives: The objectives of this experiment are:

- To learn about 7-segment display.
- To learn about alphanumeric display.
- To learn about LCD display.
- To learn about LED matrix display.

Seven segment Display

The 7-segment display, also written as "seven segment display", consists of seven LEDs (hence its name) arranged in a rectangular fashion as shown. Each of the seven LEDs is called a segment because when illuminated the segment forms part of a numerical digit (both Decimal and Hex) to be displayed. An additional 8th LED is sometimes used within the same package thus allowing the indication of a decimal point, (DP) when two or more 7-segment displays are connected together to display numbers greater than ten.

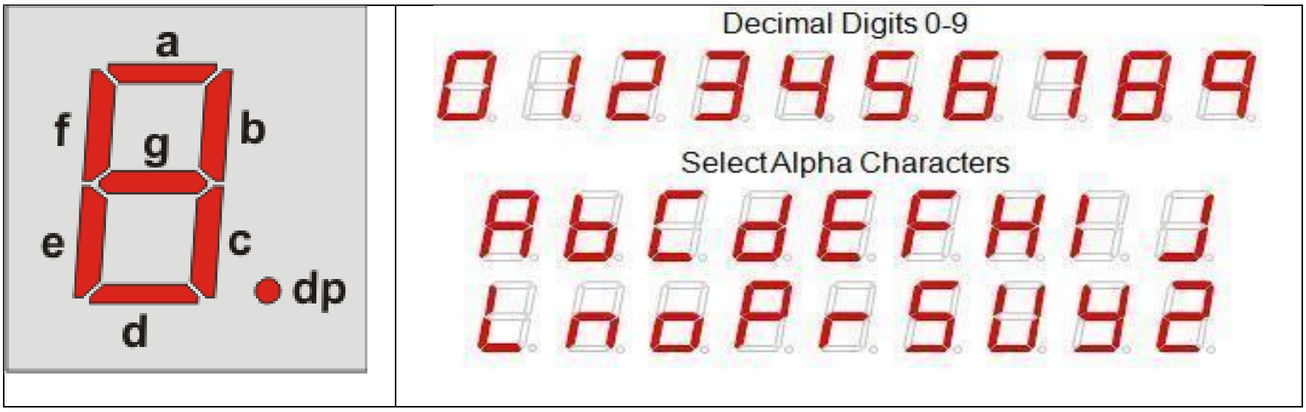


Figure 1.9: Seven segment display and pattern of digits and character

Types of 7-segment Display:

The displays common pin is generally used to identify which type of 7-segment display it is. As each LED has two connecting pins, one called the "Anode" and the other called the "Cathode", there are therefore two types of LED 7-segment display called: Common Cathode (CC) and Common Anode (CA).

The difference between the two displays, as their name suggests, is that the common cathode has all the cathodes of the 7-segments connected directly together and the common anode has all the anodes of the 7-segments connected together and is illuminated as follows.

The Common Cathode (CC) – In the common cathode display, all the cathode connections of the LED segments are joined together to logic “0” or ground. The individual segments are illuminated by application of a “HIGH”, or logic “1” signal via a current limiting resistor to forward bias the individual Anode terminals (a-g).

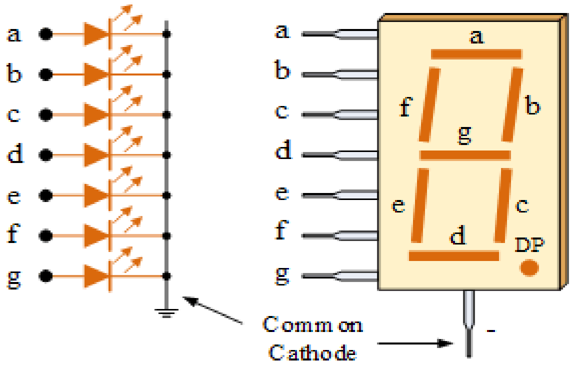
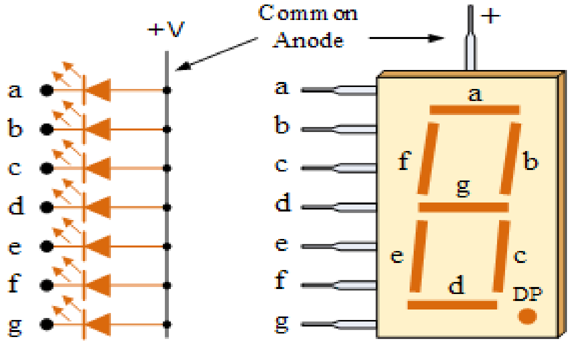


Figure 2.0: Common Cathode 7-segment display.

The Common Anode (CA) – In the common anode display, all the anode connections of the LED segments are joined together to logic “1”. The individual segments are illuminated by applying a ground, logic “0” or “LOW” signal via a suitable current limiting resistor to the



Cathode of the particular segment (a-g).

Figure 2.1: Common Anode 7-segment display.

In general, common anode displays are more popular as many logic circuits can sink more current than they can source.

Displaying in 7-segment display

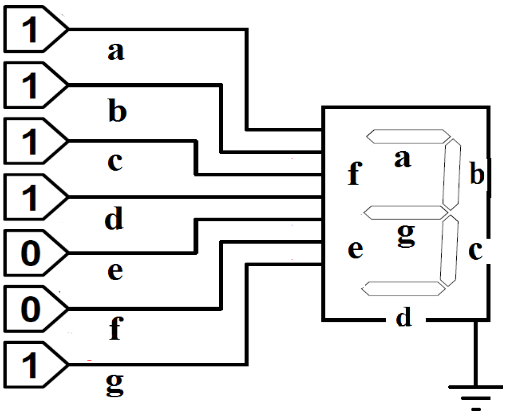


Figure 2.2 (a): Displaying 3 in common cathode 7-segment display

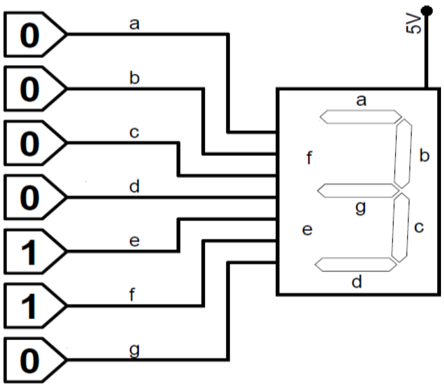
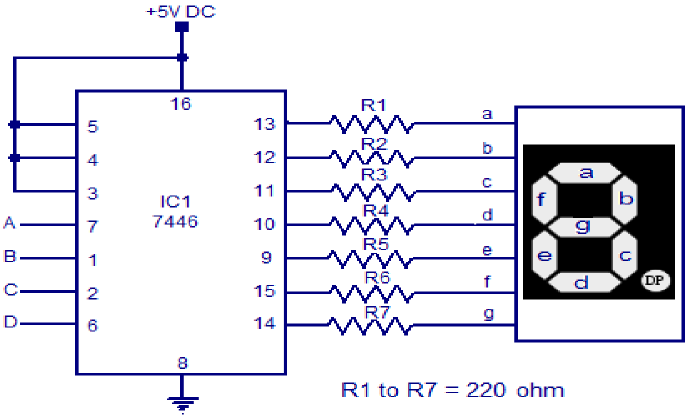


Figure 2.2 (b): Displaying 3 in common anode 7-segment display

A BCD to 7-segment decoder can be used to convert BCD number into 7-segment digit



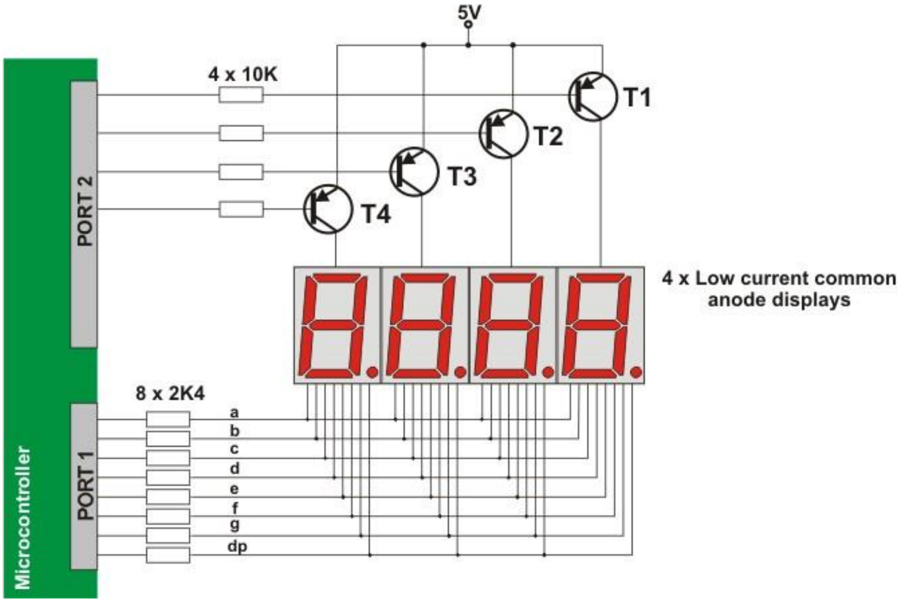
pattern.

Figure 2.3: Use of 7-segment decoder

Displaying in multiple 7-segment display

Displays connected to the microcontroller usually occupy a large number of valuable I/O pins, which can be a big problem especially if it is needed to display multi digit numbers. The problem is more than obvious if, for example, it is needed to display two 6-digit numbers (a simple calculation shows that 96 output pins are needed in this case). The solution to this problem is called MULTIPLEXING. Only one digit is active at a time, but they change their state so quickly making impression that all digits of a number are simultaneously active.

First a byte representing units is applied on a microcontroller port and a transistor T1 is activated at the same time. After a while, the transistor T1 is turned off, a byte representing tens is applied on a port and a transistor T2 is activated. This process is being cyclically



repeated at high speed for all digits and corresponding transistors.

Figure 2.4: Multiplexing technique to show number in multiple seven segment display

Pin diagram of 7-segment display

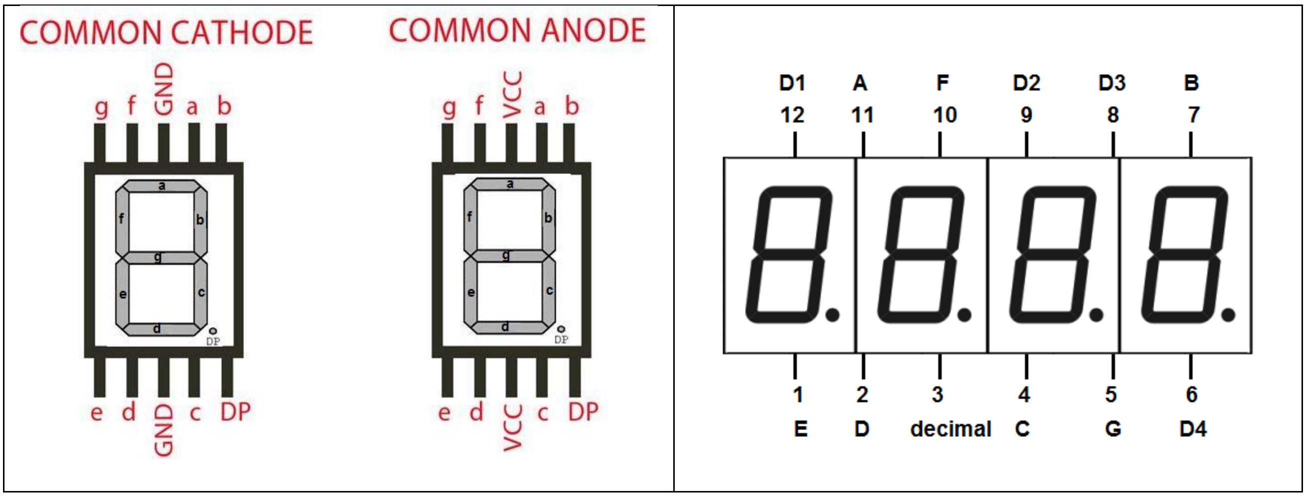


Figure 2.5: Pin diagram of 7-segment display

Alphanumeric display

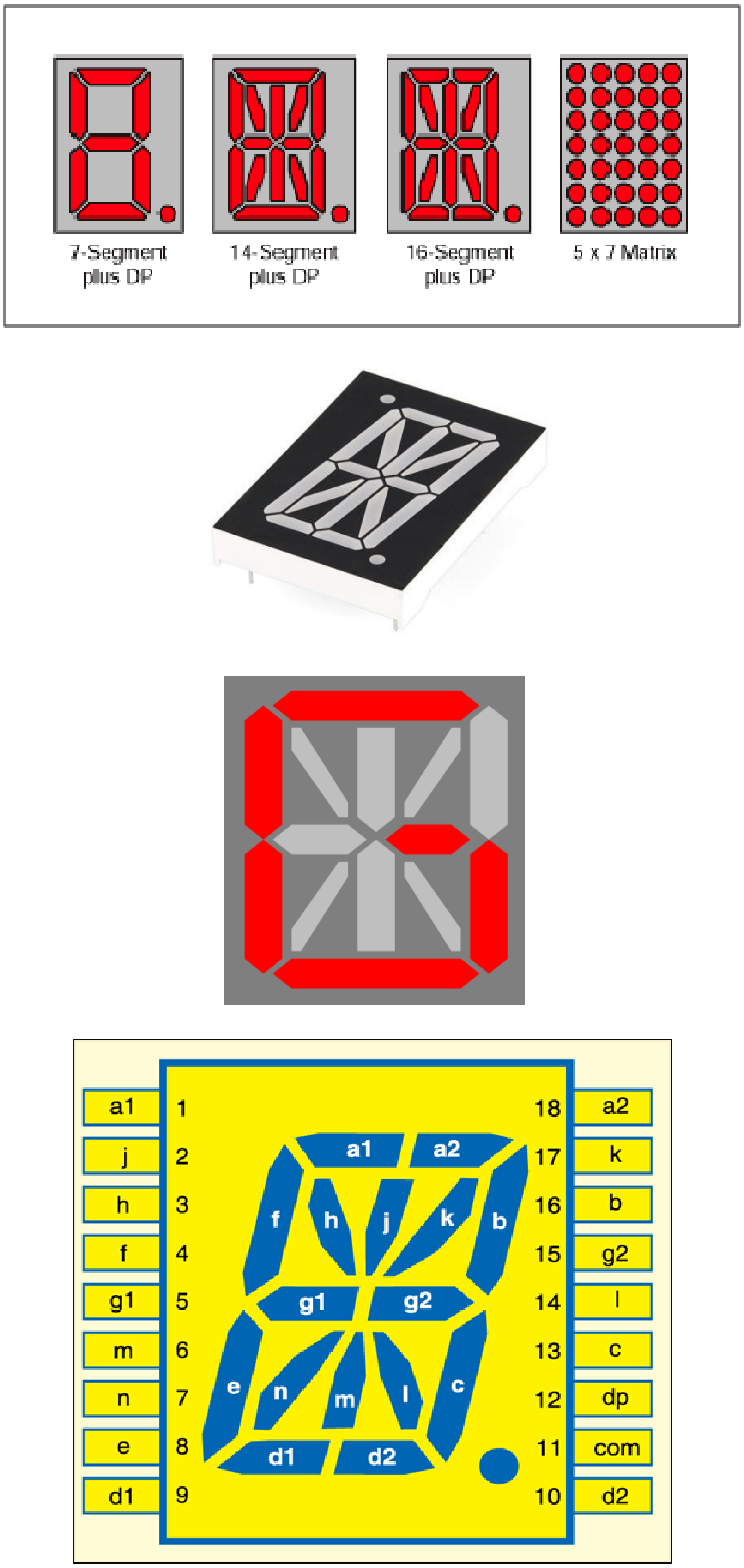


Figure 2.6: Alphanumeric display

About LCD Display:

This component is specialized to be used with the microcontrollers, which means that it cannot be activated by standard IC circuits. It is used for displaying different messages on a miniature liquid crystal display. A model described here is for its low price and great capabilities most frequently used in practice. It is based on the HD44780 microcontroller (Hitachi) and can display messages in two lines with 16 characters each. It displays all letters of alphabet, greek letters, punctuation marks, mathematical symbols etc. In addition, it is possible to display symbols made up by the user. Other useful features include automatic message shift (left and right), cursor appearance, LED backlight etc.

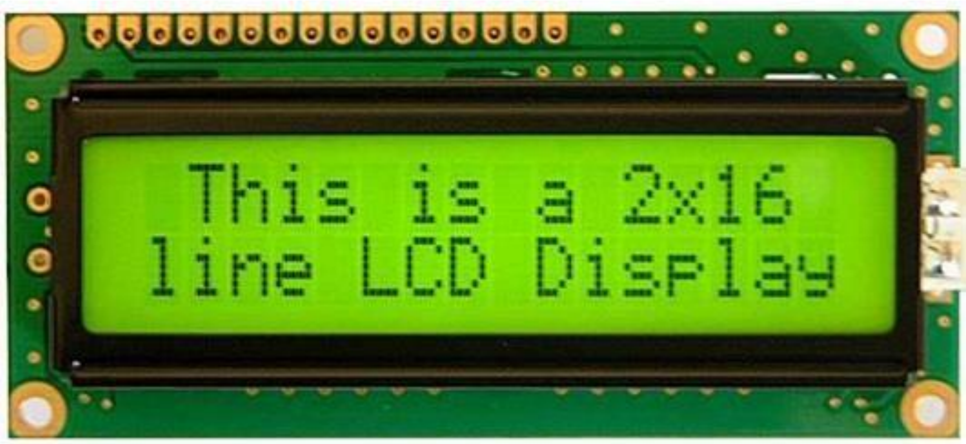


Figure 2.7: LCD Display

Along one side of a small printed board there are pins used for connecting to the microcontroller. There are in total of 14 pins marked with numbers (16 in case the backlight is built in). Their functions are described in table below:

Function	Pin Number	Name	Logic State	Description
Ground	1	Vss	-	0V
Power supply	2	Vdd	-	+5V
Contrast	3	Vee	-	0 - Vdd
Control of operating	4	RS	0 1	D0 – D7 are interpreted as commands D0 – D7 are interpreted as data
	5	R/W	0 1	Write data (from controller to LCD) Read data (from LCD to controller)
	6	E	0 1 From 1 to 0	Access to LCD disabled Normal operating Data/commands are transferred to LCD
Data / commands	7	D0	0/1	Bit 0 LSB
	8	D1	0/1	Bit 1
	9	D2	0/1	Bit 2
	10	D3	0/1	Bit 3
	11	D4	0/1	Bit 4
	12	D5	0/1	Bit 5
	13	D6	0/1	Bit 6
	14	D7	0/1	Bit 7 MSB

Table: Pin functions of 2X16 LCD display

LCD Screen

LCD screen consists of two lines with 16 characters each. Every character consists of 5x8 or 5x11 dot matrix. This book covers 5x8 character display, which is indeed the most commonly, used one.

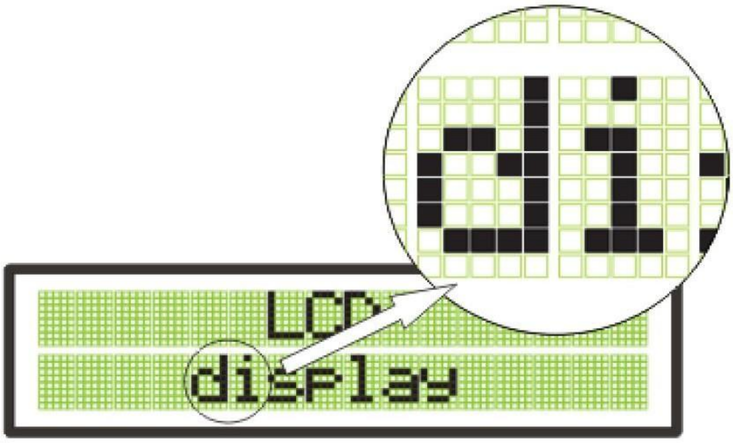


Figure 2.8: LCD screen.

Display contrast depends on power supply voltage. For that reason, varying voltage 0-Vdd is applied on the pin marked as Vee. Trimmer potentiometer is usually used for that purpose. Some LCD displays have built in backlight (blue or green diodes). When used during operation, a current limiting resistor should be serially connected to one of the pins for backlight (similar to LED diodes).

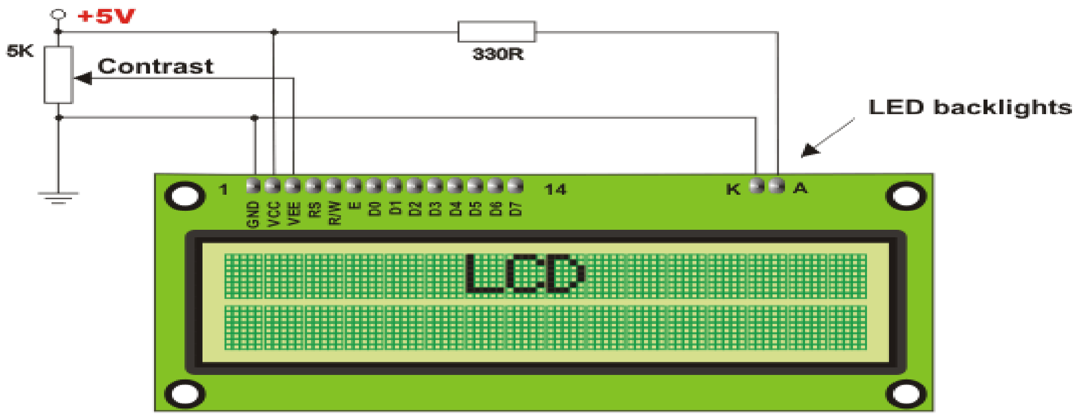


Figure 2.9: LCD contrast adjustment.

LCD Memory

LCD display contains three memory blocks:

DDRAM-Display Data RAM

CGRAM-Character Generator RAM

CGROM- Character Generator ROM

DDRAM Memory: DDRAM memory is used for storing characters that should be displayed. The size of this memory is sufficient for storing 80 characters. Some memory locations are directly connected to the characters on display.

All works quite simply: it is enough to configure display to increment addresses automatically (shift right) and set starting address for the message that should be displayed (for example 00 hex).

After that, all characters sent through lines D0-D7 will be displayed as a message. In this very case, displaying starts from the first field of the first line because the address is 00 hex. If more than 16 characters are sent then all of them will be memorized, but only first sixteen characters will be visible. In order to display the rest of them, a shift command should be used. Virtually, everything looks as if LCD display is a window which shifts left-right over memory locations containing different characters. In reality, that is how the effect of message shifting on the screen has been made.

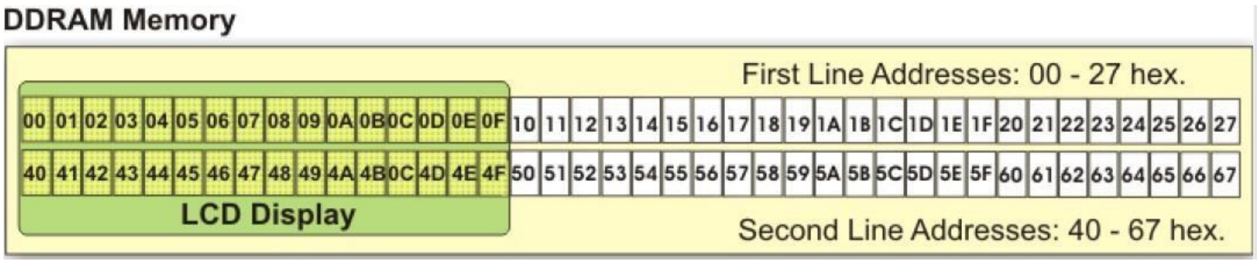


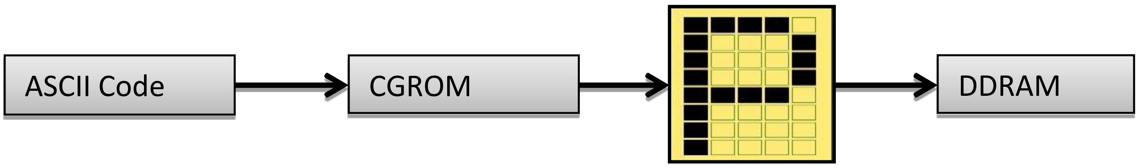
Figure 2.10: memory address of DDRAM memory

If cursor is on, it appears at location which is currently addressed. In other words, when a character appears at cursor position, it will automatically move to the next addressed location. This is a sort of RAM memory so data can be written to and read from it, but its contents is irretrievably lost upon the power goes off.

CGROM Memory: CGROM memory contains default character map with all characters that can be displayed on the screen. Each character is assigned to one memory location.

Table 2.1: CGROM memory address for different character

Addresses of CGROM memory locations match the characters of ASCII. If the program being currently executed encounters a command “send character P to port” then binary value 0101 0000 appears on the port. This value is ASCII equivalent to the character P. It is further written to LCD, which results in displaying the symbol from the 0101 0000 location of CGROM. In other words, the character “P” is displayed. This applies to all letters of the alphabet (capital and small), but not to the numbers!



Character Pattern

Figure 2.11: Process of displaying ASCII character.

As seen on the previous map, addresses of all digits are pushed forward by 48 in relative to their values (digit 0 address is 48, digit 1 address is 49, digit 2 address is 50 etc.). Accordingly, in order to display digits correctly it is necessary to add a decimal number 48 to each of them prior to sending them to LCD.

Modification of character map is not possible in CGROM.

CGRAM: This memory works same as CG ROM but as this is RAM we can modify its content any time. So, this is the place where when have to first store our custom character pattern, then that pattern can be sent to display.

The HD44780 has total 8 CG RAM memory locations. So, we can generate only up to 8 custom characters. But you can always change the content of CG RAM on the fly to generate new characters. The addresses of 8 CG RAM locations go from 0x00 to 0x07.

LCD Connection

Depending on how many lines are used for connecting LCD to the microcontroller, there are 8-bit and 4-bit LCD modes. The appropriate mode is selected at the beginning of the operation in that process called initialization. 8-bit LCD mode uses outputs D0-D7 to transfer data as explained previous.

The main purpose of 4-bit LED mode is to save valuable I/O pins of the microcontroller. Only 4 higher bits (D4-D7) are used for communication, while others may be unconnected. Each data is sent to LCD in two steps- four higher bits are sent first (normally through the lines D4-D7) and four lower bits are sent afterwards. Initialization enables LCD to link and interpret received bits correctly.

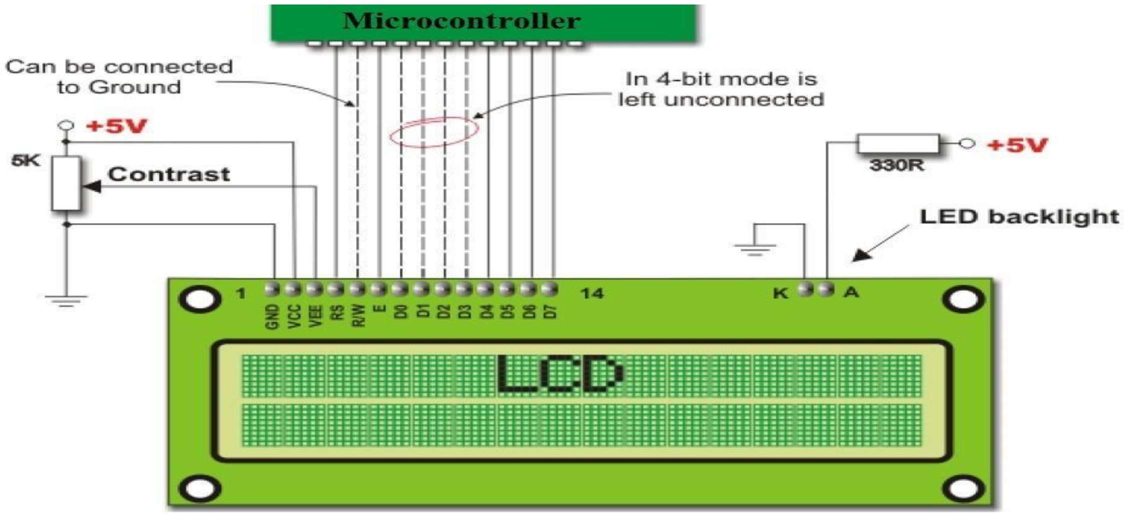


Figure 2.12: LCD connection with microcontroller.

Besides, data is rarely read from LCD (it is mainly transferred from the microcontroller to LCD) so it is often possible to save an extra I/O pin by simple connecting R/W pin to the Ground. Such saving has its price. Messages will be normally displayed, but it will not be possible to read busy flag since it is not possible to read display as well. Fortunately, there is a simple solution. After sending a character or a command it is important to give LCD enough time to do its job. Owing to the fact that execution of the slowest command lasts for approximately 1.64mS, it will be fairly enough to wait approximately 2mS for LCD.

Dot Matrix Display

In a dot matrix display, multiple LEDs are wired together in rows and columns. This is done to minimize the number of pins required to drive them. For example, a 8×8 matrix of LEDs (shown below) would need 64 I/O pins, one for each LED pixel. By wiring all the anodes together in rows (R1 through R8), and cathodes in columns (C1 through C8), the required number of I/O pins is reduced to

16. Each LED is addressed by its row and column number. In the figure below, if R4 is pulled high

and C3 is pulled low, the LED in fourth row and third column will be turned on. Characters can be displayed by fast scanning of either rows or columns.

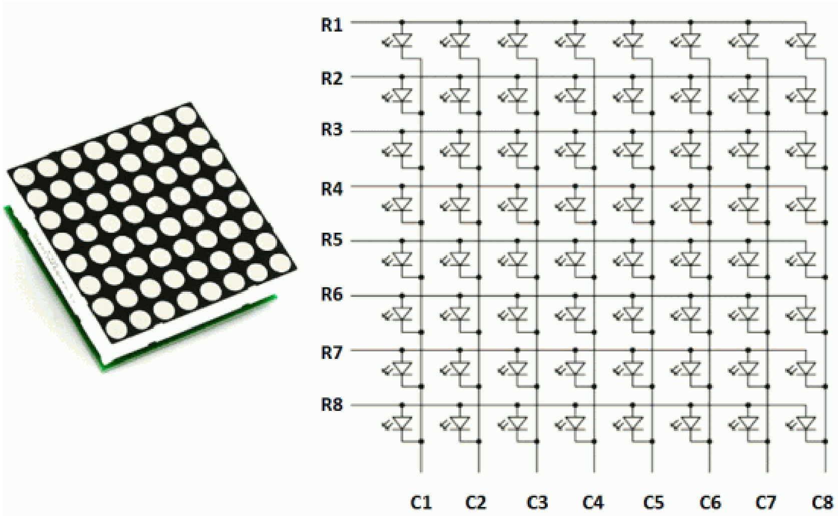
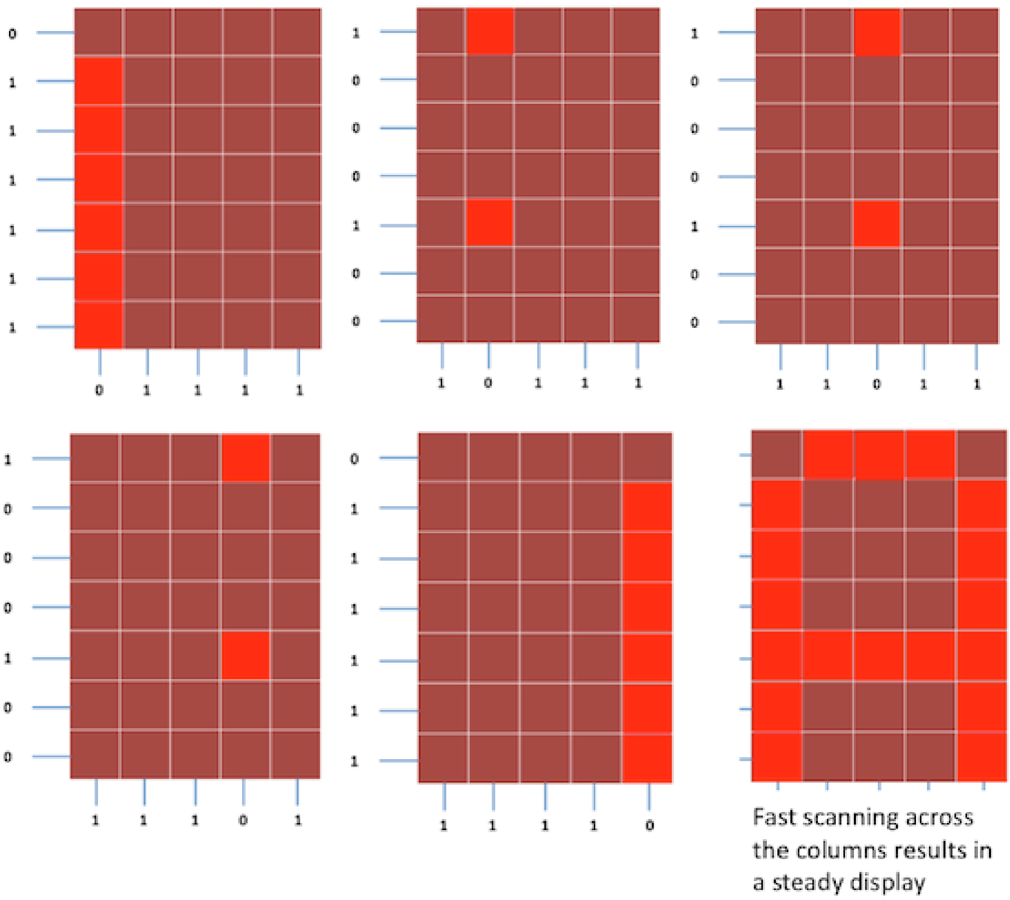


Figure 2.13: Dot Matrix Display

Now, we will learn how to display still characters by column scanning in a standard 5×7-pixel format. Suppose, we want to display the alphabet A. We will first select the column C1 (which means C1 is pulled low in this case), and deselect other columns by blocking their ground paths (one way of doing that is by pulling C2 through C5 pins to logic high). Now, the first column is active, and you need to turn on the LEDs in the rows R2 through R7 of this column, which can be done by applying forward bias voltages to these rows.



Next, select the column C2 (and deselect all other columns), and apply forward bias to R1 and R5, and so on. Therefore, by scanning across the column quickly (> 100 times per second), and turning on the respective LEDs in each row of that column, the persistence of vision comes in to play, and we perceive the display image as still.

Experiment- 6: Distance measurement by Ultrasonic Sensor.

Objective: To measure the distance by ultrasonic sensor and display it to LCD.

How Ultrasonic Sensor Works

The ultrasonic sonar sensor has four pins Vcc, Trigger, Echo and Ground. When a low to high pulse is applied to the trigger pin, an ultrasonic sound is transmitted from the transmitter. If there exist an obstacle in its range, the sound must be reflected and will be received by receiver. When the reflected sound is received the echo, pin becomes high.

The difference between times of transmitting sound (t1) and receiving sound (t2) depends on distance of object. If the time difference between time is t=t2-t1, we can write

$2d = V \times t$

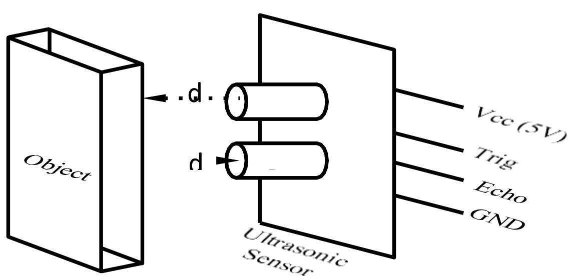


Figure-2.14 Ultrasonic Sensor

Experiment-7:

Name of the Experiment: Arduino project-I (Blinking LED light)

Components Required

- 1 × Breadboard
- 1 × Arduino Uno R3
- 1 × LED
- 1 × 330Ω Resistor
- 2 × Jumper
- USB loader For loading code.
- Arduino IDE.

Procedure

Follow the circuit diagram and hook up the components on the breadboard as shown in the image given below.

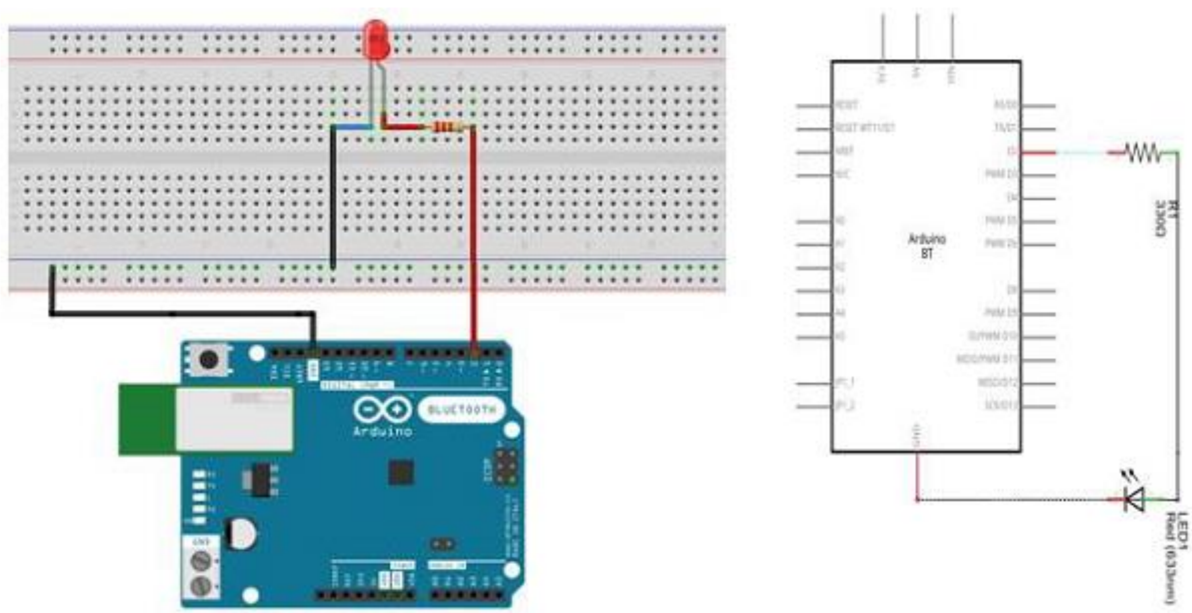


Fig-2.15: Arduino Blinking LED project.

Open the Arduino IDE software on your computer. Coding in the Arduino language will control your circuit. Open the new sketch File by clicking New.

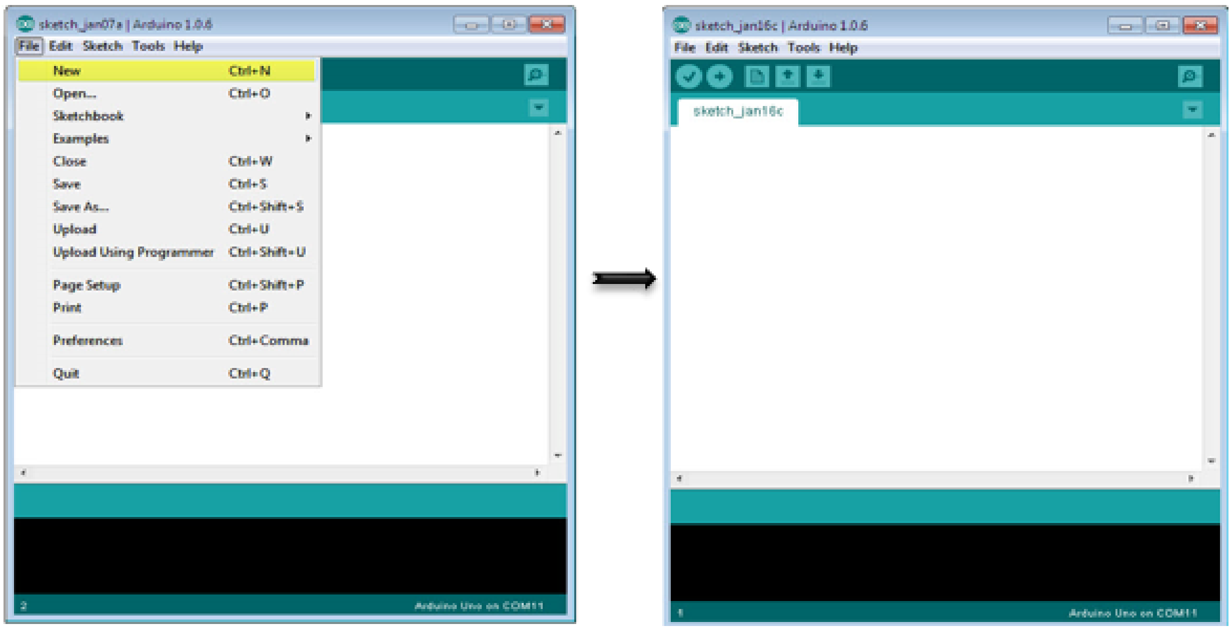


Fig-2.16: Browsing Arduino IDE.

```
/*
  Blink
  Turns on an LED on for one second, then off for one second, repeatedly.
*/

// the setup function runs once when you press reset or power the board

void setup()
{  // initialize digital pin 13 as an output.
  pinMode(2, OUTPUT);
}

// the loop function runs over and over again forever

void loop() {
  digitalWrite(2, HIGH); // turn the LED on (HIGH is the voltage level)
  delay(1000); // wait for a second
  digitalWrite(2, LOW); // turn the LED off by making the voltage LOW
  delay(1000); // wait for a second
```

```
}
```

Code to Note

pinMode(2, OUTPUT) − Before you can use one of Arduino’s pins, you need to tell Arduino Uno R3 whether it is an INPUT or OUTPUT. We use a built-in “function” called pinMode() to do this.

digitalWrite(2, HIGH) − When you are using a pin as an OUTPUT, you can command it to be HIGH (output 5 volts), or LOW (output 0 volts).

Result

You should see your LED turn on and off

Experiment-8:

Name of the experiment: Arduino with Simulation Software. (Blinking LED Traffic lights)

Software Used: TinkedCad

Equipments:

1. Resistor(220ohms)
2. Arduino Uno R3.
3. Three LED's. (RGB)

Procedure:

1. Go to TinkerCad.com for performing this simulation.
2. Go to components and select LED and Resistances.
3. Do the simulation according to the instructor.

Simulation:

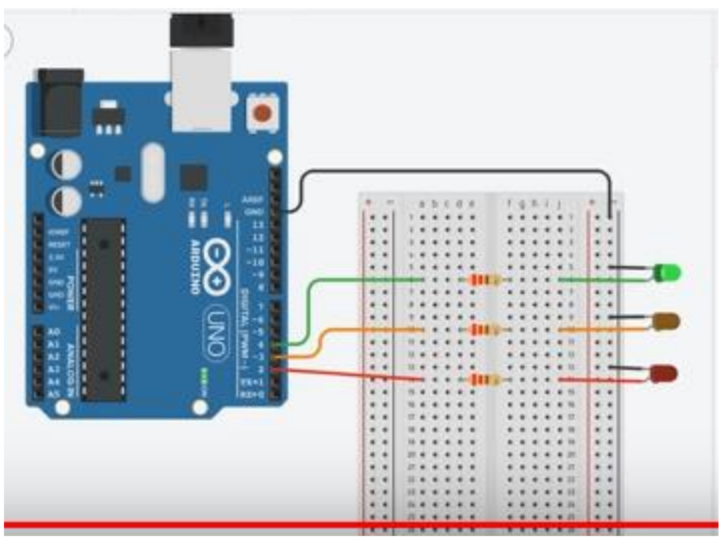
For this procedure a code is needed to be uploaded into Arduino Uno R3.

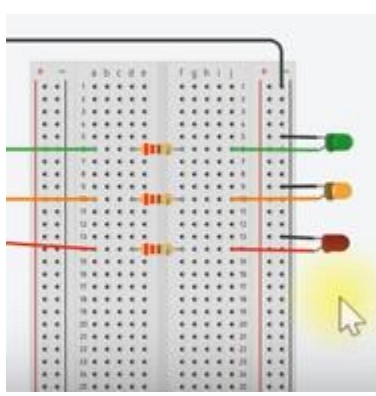
CODE:

```
void setup()

{

pinMode(4, OUTPUT);

pinMode(3, OUTPUT);

 pinMode(2, OUTPUT);

}

 void loop()

{

digitalWrite(4, HIGH); // For green LED
```

```
 delay(3000); // Wait for 3 second
 digitalWrite(4, LOW);
digitalWrite(3, HIGH); // For orange LED
delay(1000); // Wait for 1 second
 digitalWrite(3, LOW);
digitalWrite(2, HIGH); // For the red LED
delay(3000); // Wait for 3 second
digitalWrite(2, LOW);
 }
```

Simulation Output:





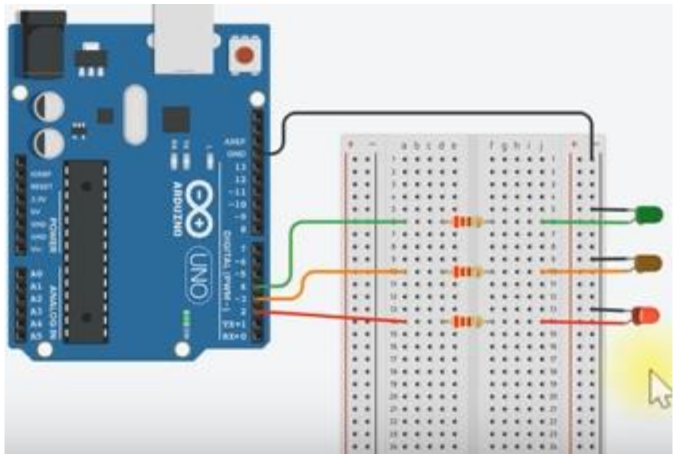


Fig-2.17: Simulated Circuit on TinkerCad.

Experiment No-09

Name of the Experiment: Circuit Simulation Using Proteus

Objective: To simulate complex circuit in convenient manner in order to find out circuit parameters.

Apparatus Name:

- Current source.
- DC voltmeter.
- DC Ammeter.
- Resistors.
- Wires.
- GND
- Battery
- Proteus

Procedure:

- Make a circuit shown in the following figure as per the instructor’s instruction
- Connect DC voltmeter in parallel and DC ammeter in series to get the desired results.
- Connect the GND else the simulation will not start.
- Now simulate to see the result.

Circuit Diagram:

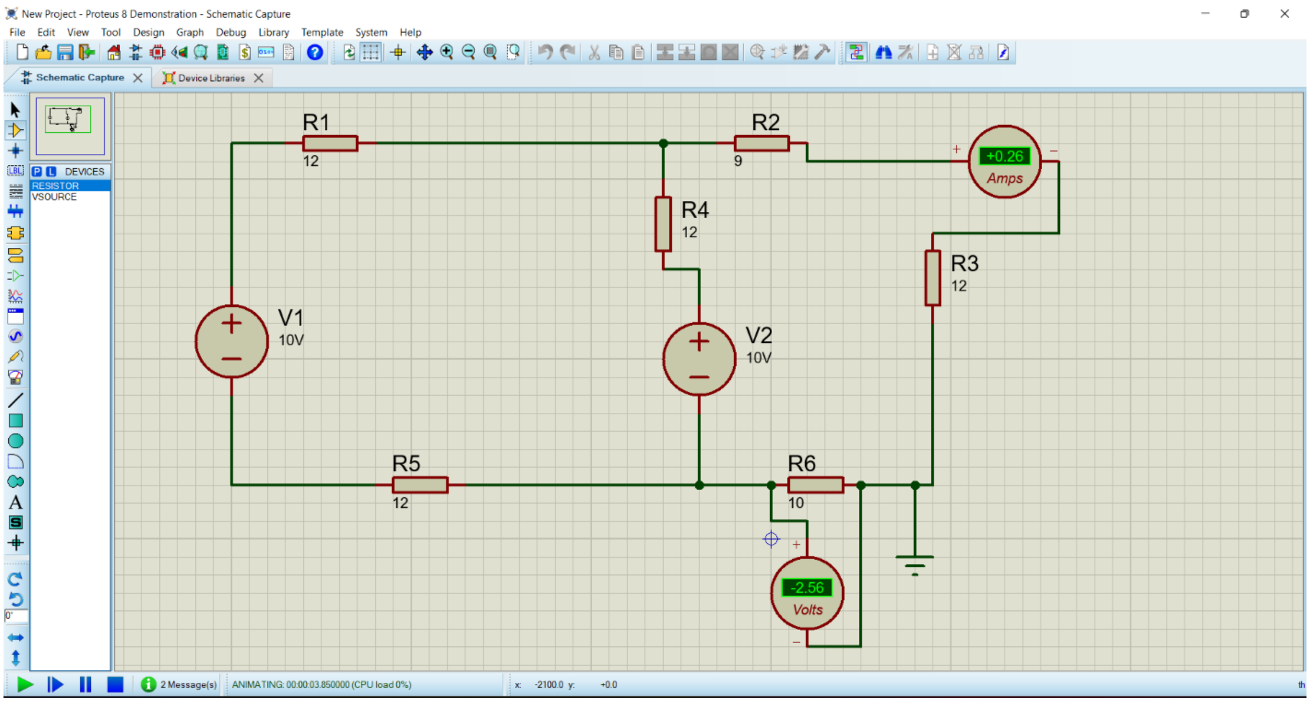


Fig-2.18: Simulation of basic circuits using resistors and ammeter, voltmeter.

Result: Desired result can be found in the DC voltmeter and ammeter easily. These values can also be found using different circuit analysis method which has been learnt in the theory course of Basic Electrical Engineering.

LAB PROJECT

Objective: Students will have to make a Project Proposal which is to be submitted to the course instructor. Course Instructor will choose one project and then the student or group of students will have to submit the project as the final project within due date.

Some Example Projects can be:

- Traffic light control with Arduino and LCD.
- Making Line follower Robot.
- Water level Control.
- Making App to control things with Arduino.